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MILITARY SPECIFICATION

CARTRIDGE, INITIATOR, M73

1. SCOPE

1.1 This specification covers Cartridge, Initiator, M73 for use with propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-A-2550 —Ammunition and Special Weapons, General Specification for
MIL-E-5272 —Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-I-45208 —Inspection System Requirements

STANDARDS

MILITARY

MIL-STD-105 —Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 —Quality Assurance Terms and Definitions
MIL-STD-639 —Surface Roughness Values Preferred for Army Ordnance Materiel

DRAWINGS

MUNITIONS COMMAND

D8594157 —Cartridge, Initiator, M73, Assembly
C8797375 —Box, Packing, Ammunition, for Cartridge, Initiator or Thruster, M38, or Cartridge, Initiator M73
C8797376 —Carton, Packing, Ammunition, for Cartridge, Initiator or Thruster, M38 or Cartridge, Initiator, M73
IEL-8594157 —Inspection Equipment List for Cartridge, Initiator, M73

PUBLICATIONS

Frankford Arsenal Manual for Functional Tests—Cartridge Actuated Devices for Aircraft Ejection Systems.
Frankford Arsenal Report No. MR-540

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **General.** Materials, parts, assemblies, chemical and physical requirements shall conform to the applicable drawings and referenced specifications.

3.2 **Initial production Sample.** Unless

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otherwise directed by the contracting officer, an initial production sample shall be required.

3.3 **Protective Finish.** When protective finish is required by component details, finish requirements shall be as required on the drawing and referenced specification and shall apply to all surfaces of the assembled cartridge regardless of contour configuration.

3.4 **Primer.** The 72M primer shall be used. The primer shall conform to the applicable drawing and referenced specification. Prior to assembly into a cartridge the primer shall be visually examined and X-rayed.

3.5 **Cartridge Assembly.** Components shall be combined to produce an assembly which conforms to the requirements of Drawing D8594157.

3.5.1 **Vibration.** The assembled cartridge shall withstand vibration prescribed in MIL–E–5272, Procedure XII, without evidence of any disintegration of the primer, igniter or propellant or derangement of components.

3.6 **Functional Requirements.**

3.6.1 The cartridge shall not fail to fire.

3.6.2 Powder Charges. The igniter and propellant charges shall be fixed by the contractor to produce the specified ballistic requirements (see 6.1).

3.6.2.1 *Primer Sensitivity.* The sensitivity of the primer in the primed cartridge shall be such that: The average critical height ($\overline{H}$) plus five standard deviations (5S) shall not exceed 15 inches. The average critical height ($\overline{H}$) minus two standard deviations (2S) shall not be less than 2½ inches.

3.6.3 *Functioning.* The cartridge when tested in an M3A1 Initiator shall produce the pressures and ignition delay at the temperatures specified in table I.

3.7 **Workmanship.** The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

TABLE I

Temperature Degree (°) Fahrenheit (F.)	Pressure Pounds Per Square Inch (psi) Minimum		Ignition Delay Milliseconds, Max. (See 6.2)
	Min.	*Max.*	
−65° to −70° (F)	550	1450	65
65° to 75° (F)	700	1550	65
160° to 165° (F)	850	1650	65

3.7.1 *Metal Defects.* The metal part shall be free from cracks, splits, laminations, perforations, porosity or any similar defect.

3.7.2 *Burr.* The part shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.7.3 *Foreign Matter.* The part or assembly shall not contain chips, dirt, grease, oil, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *Quality assurance terms and definitions.* Reference shall be made to MIL–STD–109 to define the quality assurance terms used.

4.1.2 *Inspection.* Inspection shall be in accordance with MIL–A–2550 and MIL–I–45208, except as specified herein.

4.2 **Initial production sample.** Prior to the initiation of regular production, the contractor shall submit an initial production sample

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to a Government approved facility for evaluation. The sample shall consist of 130 cartridges and 10 sets of unassembled components including explosive components. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing must be the same as will be used in regular production and shall be obtained from the same source of supply. The contractor shall inspect the sample in accordance with his inspection system and other inspection requirements of MIL–I–45208. After inspection and provisional acceptance at source, the initial production sample shall be inspected at the Government approved facility for all requirements of the drawings and specifications. The Government reserves the right to require new initial production samples until such time as an acceptable sample is submitted.

4.3 Inspection Provisions.

4.3.1 *Lot.*

4.3.1.1 *Submission of Product.* The product shall be submitted in accordance with MIL–STD–105.

4.3.1.2 *Size of Lot.* Unless otherwise specified by the procuring agency, an ammunition lot shall consist of 1200 cartridges, maximum, exclusive of test samples and shall consist of:

(a) Components produced by a homogeneous process
(b) Propellant from not more than one lot
(c) Igniter from not more than one lot
(d) Primers from not more than one lot

4.3.2 *Examination.* One hundred per cent examination shall be performed for all critical defects. Examination for major or minor defects shall be performed on a defect basis in accordance with the classification of defects, using the sampling plans and criteria in tables II and III. For an individual defect (single characteristic) the criteria in table II shall be used. For total defects within a major or minor group, the criteria in table III shall be used. All nonconforming material shall be rejected. Four power (4X) magnifications may be used in performing visual examination for such defects as burr and foreign matter. The Ordnance Standard Finish Blocks prescribed in MIL–STD–639 shall be used as a basis for comparison for surface roughness determination.

TABLE II

Lot Size	Sample Size	Acceptance Numbers	
		Major	Minor
1– 300	100% Examination	0	0
301–1300	110	1	2
1301–3200	150	2	3

TABLE III

Number of defects in class	*Total major defects allowable*
2– 3	1
4– 7	2
8–19	3

Number of defects in class	*Total minor defects allowable*
2– 3	2
4– 7	3
8–16	4

4.3.2.1 *Classification of Defects.*

4.3.2.1.1 *Cartridge, Initiator, M73, Assembly, Drawing D8594157.*

Defect	*Method of inspection*
CRITICAL	
1. Depth to primer pocket face	Gage
2. Primer seating depth	Gage
3. Any part missing	X-Ray/Visual [1]
4. Improper assembly	X-Ray/Visual [1]
5. Mixed ammunition	Visual
6. Metal defective	Visual
7. Crimp missing or in correct	Visual
8. Marking missing, incorrect or illegible	Visual

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Defect	Method of inspection
MAJOR	
101. Cartridge profile failure	Gage
102. Diameter of body, max.	Gage
103. Diameter of head, max.	Gage
104. Length to datum diameter, max.	Gage
105. Protective finish inadequate or defective	Visual
106. Foreign matter	Visual
107. Burr	Visual
108. Primer crimp missing	Visual
MINOR	
201 Total length, Max.	Gage
202. Dent	Visual
203. Draw scratch	Visual
204. Fold, wrinkle, buckle or bulge	Visual
205. Surface finish improper	Visual
206. Varnish inadequate or missing	Visual

[1] The assembled cartridge shall be X-rayed with the long axis of the cartridge perpendicular to the rays of the X-ray machine. The X-ray negatives, properly identified, shall be forwarded to the procuring activity.

4.3.2.1.2 *Carton, Packing, Ammunition, Drawing C8797376.*

Defect	Method of inspection
MAJOR	
None	
MINOR	
201. Quantity of contents incorrect	Visual
202. Contents loose	Manual/Visual
203. Separator or filler missing	Visual
204. Heat sealer incomplete, loose or distorted	Visual
205. Cut or damage through impregnated box	Visual
206. Marking missing, incorrect or illegible	Visual

4.3.2.1.3 *Box, Packing, Ammunition, Drawing C8797375.*

MAJOR	
None	
MINOR	
201. Contents loose	Manual/Visual
202. Hardware, missing, loose, broken or improperly engaged	Visual
203. Strapping missing, loose, broken or improperly assembled	Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of cartons incorrect	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

4.3.2.2 *Primer Examination.* Prior to assembly into a cartridge, each primer shall be visually examined for any discoloration of the foiling paper or a malformation of any kind. Any discoloration of the foiling paper or malformation or failure to comply with the requirements of the applicable drawing and referenced specification shall be cause for rejection of the primer.

4.3.3 *Inspection Testing.* The following tests shall be performed in accordance with the provisions of 4.4, except as provided in MIL–I–45208.

4.3.3.1 *Protective Finish.* The number of units for protective finish tests shall be in accordance with the applicable specifications. Failure of any sample to comply with the requirements of the applicable specifications shall be cause for rejection of the lot.

4.3.3.2 *Primer X-Ray Inspection.* Each primer shall be X-rayed. Failure of any primer to comply with the applicable requirements of 3.4 shall be cause for rejection of the primer.

4.3.3.2.1 *Primer Sensitivity.* A sample of 100 primed cartridges from the initial production sample and 50 primed cartridges from each subsequent production lot shall be tested for sensitivity. Failure to comply with the requirements of 3.6.2.1 shall be cause for rejection of the lot.

4.3.3.3 *Airtightness.* Each cartridge shall be subjected to the specified pressure differential. Failure to comply with the requirements of the drawing shall be cause for rejection of the cartridge.

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4.3.3.4 *Vibration Test (Initial production Sample).* Twelve of the 30 assembled, X-rayed cartridges of the initial production sample, properly identified, shall be subjected to vibration as directed in 3.5.1. After the vibration test the 12 vibrated cartridges shall be X-rayed as directed in Note ' 4.3.2.1.1. Evidence of any disintegration of the primer, propellant or igniter or derangement of any component shall be cause for rejection of the lot.

4.3.3.5 *Vibration Test (Production Lot Sample).* When it becomes necessary to make a change in the lot of primers or propellant or igniter after production is initiated, 30 cartridges from the first production lot thereafter shall be tested in accordance with 4.3.3.4.

4.3.3.6 *Functioning.* The 30 test cartridges, including the 12 vibrated cartridges, of the initial production sample (or subsequent sample of 30 cartridges when a change in primers, propellant or igniter has been made) and 15 cartridges from each subsequent lot shall be temperature conditioned and functionally tested. Failure of any cartridge to comply with the requirements of 3.6 shall be cause for rejection of the lot.

4.3.4 *Inspection Equipment.* The examination and tests shall be made using equipment listed on IEL–8594157 except as provided in MIL–I–45208.

4.4 Test Methods and Procedures.

4.4.1 *Protective Finish.* Test procedures for protective finish shall be in accordance with the applicable specification referenced on the drawing.

4.4.2 *Primer X-Ray.* Prior to assembly into a cartridge, the primer shall be X-rayed with the side of the cup perpendicular to the rays to determine compliance with the requirements of 3.4. The X-ray negatives, properly identified, shall be forwarded to the procuring activity.

4.4.2.1 *Primer Sensitivity.* Remove and discard all powder charges from the test cartridges by sectioning the cartridge case around its periphery ½ inch, maximum, above the base of the cartridge. Drop test one half of the prescribed sample at each of two heights in accordance with the method prescribed in Frankford Arsenal Report No. MR–540 using a primer sensitivity test machine with a cartridge holder approved by the procuring activity. Tests shall be made with a drop test firing pin, whose pin contour radius shall be .078 minus .002 inches, and the drop weight shall be a steel ball, 1 3/16 inches in diameter, weighing 3.94 plus or minus .02 ounces. All firing pin indent centers shall lie within .020 inches of the center of the primer. The two drop test heights may be determined by analysis of the drop test data obtained during the acceptance testing of the primer. The two drop test heights selected shall be such that all requirements specified in Frankford Arsenal Report No. MR–540 are met. Compute $\overline{H}$ and S in accordance with referenced report, then compute $\overline{H}$ plus 5S and H minute 2S. Record data on Form No. ORDBA–770.

4.4.3 *Airtightness.* The method for determining airtightness shall be by use of equipment approved by the procuring activity.

4.4.4 *Vibration.* The method and procedure for performing the vibration test shall be in accordance with MIL–E–5272, Procedure XII.

4.4.5 *Functional Tests.* The following tests will be performed by the Government:

4.4.5.1 *Initial production Sample.* Cartridges of the initial production sample shall be grouped and temperature conditioned in accordance with 4.4.5.3 as follows: 10 cartridges conditioned at minus 65 degrees F. to minus 70 degrees F.; 10 cartridges at plus 65 degrees F. to plus 75 degrees F.; 10 cartridges at plus 160 degrees F. to plus 165 degrees F., and then tested in accordance with 4.4.5.4. Each group of 10 cartridges shall include 4 of the 12 cartridges subjected to the vibration test required in 4.3.3.4.

4.4.5.2 *Production Lot Sample.* Cartridges of the production lot sample shall be grouped,

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temperature conditioned in accordance with 4.4.5.3 as follows: 5 cartridges conditioned at minus 65 degrees F. to minus 70 degrees F.; 5 cartridges at plus 65 degrees F. to plus 75 degrees F.; 5 cartridges at plus 160 degrees F. to plus 165 degrees F., and then tested in accordance with 4.4.5.4.

4.4.5.3 *Temperature Conditioning.* Cartridges shall be loaded into acceptable M3A1 Initiators with end caps to covers the gas ports during the conditioning period. The temperature of the propellant within the cartridge shall be considered the conditioning temperature and will be determined by means of a thermocouple attached to the center of the propellant charge of a specimen cartridge assembled in a standard test initiator device similar to the initiators undergoing tests. The conditioning time shall be a minimum of one hour longer than the time necessary to bring the element contained in the specimen cartridge to the required temperature.

4.4.5.4 *Test Procedures.* Upon completion of the required temperature conditioning the sample shall be ballistically tested in accordance with "Frankford Arsenal Manual of Functional Tests for Cartridge Actuated Devices for Aircraft Ejection Systems" (or such method as is approved by the contracting officer) within five minutes after removal from the conditioning chamber. The sample shall be reconditioned if this time is exceeded. When reconditioning samples at minus 65 degrees F. to minus 70 degrees F., all condensation shall be removed from interior surfaces of the initiators before the initiators are returned to the conditioning chamber.

5. PREPARATION FOR DELIVERY

5.1 Packing and Marking. Level A. Packing and marking shall be in accordance with Drawings C8797375 and C8797376, or as required by contract.

5.2 Data cards. Data card information shall be in accordance with Specification MIL–A–2550.

6. NOTES

6.1 Advisory Loading Data:

Black Powder (Grade A–4)	20.0 grains minus 3.0 grains
Propellant, Composite, HES 5808 Type (8 pcs.)	46.0 grains minus 6.0 grains

Source: Hercules Powder Co., Kenvil, N. J. or equal

Advisory Propellant Grain Dimensions:

Length	0.450 inches
Diameter	0.200 inches
Perforation Diameter	0.014 inches
Web thickness	0.040 inches
Number of Perforations	7

6.2 Ignition Delay is defined as the time that elapses between the firing of the primer and the beginning of a continuous pressure-rise on a pressure time record.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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